

Hostel Management System

Software Requirements Document — Phase 1 (Owner & Hostel Operations)
Prepared for freelance development tender | Version 1.0

1. Project Overview

The client owns and operates multiple hostels and requires a centralized software system to manage them. The system will be built in phases. Phase 1 focuses exclusively on backend management tools used by the company (Super Admin) and hostel owners — there is no resident-facing app in this phase.

The platform must support a single Super Admin who onboards hostel owners, and owners who independently manage their own hostels, residents (hostelers), rooms, and finances.

2. Technology Stack (Mandatory)

- Frontend (Mobile App): React Native
- Backend: Python (REST API — FastAPI or Django REST Framework, freelancer to propose)
- Database: PostgreSQL or MySQL (freelancer to propose with justification)
- Deployment: Docker and Docker Compose for all services (backend, database, and any supporting services). Must include working Dockerfile(s) and docker-compose.yml that allow the client to run the entire system locally or on a server with a single command.
- File Storage: Secure storage solution for Aadhaar card photos and other documents (e.g., S3-compatible storage or server-based storage with access control) — freelancer to propose.
- Authentication: Token-based authentication (JWT) with role-based access control (RBAC).

3. User Roles & Hierarchy

Role	Created By	Core Responsibility
Super Admin	Exists by default (one, set up during deployment)	Onboards hostel owner accounts and grants login credentials; oversees the entire platform
Owner	Super Admin	Manages one or more hostels assigned to them: hostelers, rooms, inventory/finance, notifications

Note: A single Owner account can be linked to multiple hostels (multi-hostel ownership), and the Owner can switch between hostels within the app.

4. Functional Requirements

4.1 Super Admin Module

1. Secure login for Super Admin (seeded/default account, password change enforced on first login).
2. Create, edit, and deactivate hostel records (hostel name, address, contact details, number of floors/rooms).
3. Create Owner accounts and assign one or more hostels to an Owner.
4. Auto-generate and share login credentials with Owner (or allow Super Admin to set them manually).
5. Reset Owner passwords / disable Owner access.
6. Dashboard showing all hostels, total owners, and high-level summary (total hostellers, occupancy, revenue snapshot across all hostels).
7. View activity logs of Owners (audit trail) for accountability.

4.2 Owner Module

If an Owner manages multiple hostels, all features below operate per-selected-hostel, with a hostel switcher in the app.

8. Secure login using credentials provided by Super Admin; forced password reset on first login.
9. Switch between assigned hostels (if more than one).

4.2.1 Hosteler (Resident) Management

- Add a new hosteler with full profile: name, phone number, email, permanent address, emergency contact, date of joining, photo.
- Upload and store Aadhaar card photo (front & back) securely against the hosteler profile.
- Edit hosteler details and mark a hosteler as inactive/vacated, with vacate date and reason.
- Search and filter hostelers by name, room, status (active/inactive), or joining date.
- Maintain a complete history of past hostelers per hostel (not deleted, archived).

4.2.2 Room Management

- Add rooms with room number, floor, room type (single/double/triple/dormitory), capacity, and monthly rent.
- Assign a hosteler to a specific room and bed/slot.
- Re-assign / transfer a hosteler from one room to another, with history retained.
- View room-wise occupancy: total beds, occupied, vacant, in a single dashboard view.
- Prevent over-allocation — system should not allow assigning more hostelers than room capacity.

4.2.3 Notifications & Alerts

- Automated monthly alert to the Owner showing the count of active hostelers on a given date each month (e.g., headcount snapshot on the 1st of every month, or a date configurable by the Owner).
- Alert when a room reaches full occupancy or when a hosteler's rent/payment is overdue (recommended — see Section 5).
- In-app notification center listing all alerts, with push notification support on the mobile app.

4.2.4 Inventory & Finance Management

- Record income: rent/fees collected from hostelers (amount, date, mode of payment, hosteler reference).
- Record expenses: money the Owner spends on the hostel (groceries, maintenance, staff salary, electricity/water bills, repairs, etc.) with category, amount, date, and optional receipt/photo upload.
- Monthly and custom-date-range summary report: total income vs total expense vs net profit/loss, per hostel.
- Basic inventory list for hostel assets (beds, furniture, appliances) with quantity and condition — optional but requested by client.

5. Recommended Additions (Nice-to-Have)

The client has asked the freelancer to propose any additional features that would strengthen the system. The following are suggested for consideration and should be quoted as optional additions:

- Rent due-date tracking with automatic overdue reminders per hosteler.
- Payment receipt generation (PDF) for each rent collection.
- Multiple staff sub-roles under Owner (e.g., Manager/Warden) with limited permissions, for larger hostels.
- Visitor/guest log register.
- Complaint/maintenance request tracking raised against a room.
- Export of hosteler lists, financial reports, and occupancy reports to Excel/PDF.
- Dashboard charts: occupancy trend, monthly revenue trend, expense breakdown by category.
- Document expiry tracking (e.g., ID proof renewal reminders).
- Data backup and restore mechanism.
- Future Phase 2 (not in current scope, for context only): a resident-facing app/portal for hostelers to view their room, dues, and notices.

6. Non-Functional Requirements

- Role-based access control: Super Admin and Owner must only see data they are authorized for.
- Data security: Aadhaar and personal data must be encrypted at rest and transmitted over HTTPS.
- Multi-tenancy: data of one hostel/owner must be fully isolated from another.

- Scalability: architecture should support adding new hostels and owners without redesign.
- Mobile app must work on both Android and iOS via React Native.
- API documentation (Swagger/OpenAPI) to be provided.
- All services must be containerized and run via Docker Compose with clear setup instructions (README).

7. Deliverables

10. Complete source code (frontend React Native app + Python backend) with version control (Git repository access).
11. Dockerfile(s) and docker-compose.yml for backend, database, and any supporting services.
12. Database schema/ER diagram.
13. API documentation.
14. Deployment guide / README for setting up the system from scratch.
15. APK/build for Android (and iOS build if applicable) for testing.
16. One round of bug-fixing support post-delivery (duration to be agreed in contract).

8. Out of Scope (Phase 1)

- Resident/hosteler-facing mobile app or login (planned for a future phase).
- Online payment gateway integration (can be discussed as an optional add-on quote).
- SMS/WhatsApp integration (can be discussed as an optional add-on quote).

9. Submission Requirements for Freelancers

Interested freelancers/agencies are requested to submit a proposal including:

17. Technical approach and proposed database/architecture (confirming React Native + Python + Docker stack).
18. Itemized cost and timeline, with the optional Section 5 features quoted separately.
19. Relevant past work/portfolio (similar admin-panel or multi-tenant systems preferred).
20. Proposed payment milestones.
21. Warranty/support period offered post-delivery.